

ECON 383 Proposal: The Impact of Indian Point's Closure on Electricity Prices

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- In 2021 the Indian Point Energy Center in Westchester, NY closed down
 - Reactor 2 closed in April 2020 while Reactor 3 closed in April 2021 ¹
- Indian Point produced ~ 2 GW - roughly 5% of NY State's total energy generation capacity and 25% of New York City's energy usage
 - Supporters of closure claimed that the plant had adverse impacts on wildlife and was dangerous
 - Opponents touted the plant's clean energy capacity and claimed the closure would spike electricity prices
- I analyze this claim - testing whether or not the closure of Indian Point actually did increase power prices

¹Reactor 1 closed in October 1974

Strategy

- I plan to use a difference-in-differences (DiD) strategy to estimate the effect of the closure on electricity prices^a in New York
 - DiD uses control groups to come up with the counterfactual (what if without treatment) trend for the treated group
- Focusing on three core-based statistical areas (CBSAs)
 - New York
 - Philadelphia
 - Washington D.C.
- PHL and D.C. are in the PJM electricity market while NYC is in the NYISO electricity market
 - Reasonable to assume the depth of PJM's market would largely insulate PHL and D.C. from shocks in NYISO

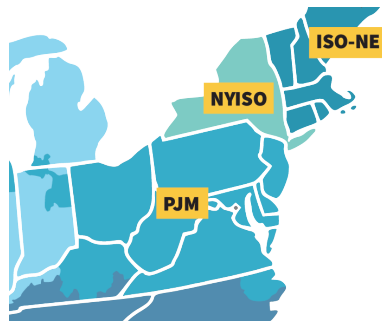


Figure: ISOs and RTOs in the Mid-Atlantic and Northeastern U.S.^b

^aSource: U.S. Energy Administration

^bSource: U.S. Energy Administration

Data and Potential Issues

- Staggered closure of Indian Point
 - Solution: Two dummies, one for each treatment event
- Exogenous variation in electricity prices via inputs
 - Solution: Control for prices that are set regionally (gas²) but not those that are global (oil, coal)
- Demand shocks that affect prices but are also partially endogenous by laws of supply and demand
 - Solution: Use lagged GDP³ to account for shocks (assuming that major factors on prices are short-run fixed)
 - Solution: Use average temperature data⁴ to account for effect of heating/AC

²Source: U.S. Energy Administration

³Source: U.S. Bureau of Economic Analysis

⁴Source: U.S. Climate Data

Data and Potential Issues cont.

- Other random exogenous shocks over time or by city
 - Fixed-effects at the city and time levels
- Inference issues arising from panel data or other arbitrary correlations
 - Solution: Heteroskedastic-robust autocorrelative (HAC) standard errors
 - Solution: Clustering of errors at city and time levels
- Clustering leads to a sample size issue ($N=3$) making asymptotics biased
 - Solution: Jackknife (re-estimate model to simulate standard errors) or Driscoll-Kraay standard errors (good for moderate N , large T samples)